

BPHARM
(SEM IV) THEORY EXAMINATION 2024-25
PHYSICAL PHARMACEUTICS II – THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

10 x 2 = 20

a.	Mention characteristics and applications of micro emulsion.
b.	Define deflocculated suspension.
c.	List the chemical factors effect drug degradation.
d.	Mention factors influencing viscosity.
e.	What do you mean by 'pseudo-zero order kinetics'?
f.	How to prevent photolytic degradation?
g.	Differentiate between true density and bulk density
h.	Define porosity. Write its applications in pharmacy.
i.	What is angle of repose and mention its importance?
j.	What is Zeta potential?

SECTION B

2. Attempt any two parts of the following:

2 x 10 = 20

a.	Explain physical and chemical factors influencing the chemical degradation of pharmaceutical products.
b.	What do you mean by Newtonian systems? Explain any one method for determination of viscosity of Newtonian systems in detail.
c.	Explain formulation of emulsions by HLB method. Discuss stability and rheological behavior of emulsions.

SECTION C

3. Attempt any five parts of the following:

7 x 5 = 35

a.	What is thixotropy and mention its importance in pharmacy.
b.	State Heckel equation and explain its pharmaceutical significance.
c.	Describe about the methods for determining surface area of particles.
d.	Explain the effect of electrolytes on colloid dispersions.
e.	Give the working principle of Andreasen apparatus with the help of a labeled diagram.
f.	Explain various flow properties of powder.
g.	Discuss formulation and evaluation of flocculated and deflocculated suspensions.